Mayukh Banerjee

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EDUCATION

Vellore Institute of Technology **Vellore, Tamil Nadu**

B.Tech in Information Technology *Expected Graduation, June 2027*

- **Concentrations:** Machine Learning, AI, Full Stack Development
- **GPA:** 8.43/10
- **Related Coursework:** Data Structures & Algorithms, Object Oriented Programming, Database Management Systems, Software Engineering, Python, Java, Operating Systems, Computer Networks

SKILLS

Programming Languages: Python, C++, C , JavaScript, Java
Libraries and Frameworks: scikit-learn, Keras, XGBoost, Hugging Face, TensorFlow, PyTorch, NLTK, NumPy, Pandas, Seaborn
Technologies: Oracle SQL, Git, GitHub, PyCharm, Selenium, FastAPI, REST APIs, Gemini API
Tools: Excel, PowerPoint, Google Apps Script, Google Workspace

PROJECT EXPERIENCE

LeBy: Intelligent Legal Assistant (RAG + Machine Learning) *VIT | December 2025*

- **Hybrid Legal AI System:** Designed a Retrieval-Augmented Generation (RAG) pipeline using FAISS and Gemini API to summarize and interpret long legal documents with grounded, **hallucination-reduced** responses.
- **Explainable ML Classification:** Implemented a TF-IDF + Logistic Regression classifier to categorize legal documents (contracts, employment, property, criminal law), enabling transparent and interpretable predictions.
- **Full-Stack Deployment:** Built a scalable FastAPI backend with session-based FAISS indexing and a React frontend for document analysis and legal Q&A, delivering structured responses in <5s per query.

AI Scheduling Agent & Voice Assistant - (Ongoing) *VIT | September 2025*

- **AI Scheduling Agent**: Architected a **multimodal agent** that orchestrates a full-stack **Speech-to-Text, LangGraph, and Text-to-Speech** pipeline to process user requests via both voice and text.
- **Agentic Tool Integration**: Engineered dynamic, timezone-aware tools for the Google Calendar API (read/write), enabling the agent to parse complex natural language commands into structured, executable API calls.
- **Full-Stack Implementation:** Built with a Python (FastAPI, LangGraph) backend and a Next.js/React frontend to deliver a dual-interface dashboard with seamless text and voice chat capabilities.

Fake News Detection & Explainable AI System *VIT | August 2025*

- **Fake News Classifier**: Built a dual-model pipeline (**Logistic Regression + Agentic AI**) achieving 92% accuracy on 40,000+ news articles.
- **Data Preprocessing & Explainability**: Cleaned and vectorized 80,000+ text samples, extracting top 5 weighted keywords per article for transparency.
- **Tech Stack & Deployment**: Implemented with Python, FastAPI, and React; deployed the intelligence to a user-friendly frontend supporting text and URL, delivering real-time results in <5s.

EXPERIENCE

Futurepreneurs 10.0 & E-Summit *E-CELL - VIT*

- **Ideation & Execution:** Conceptualized and developed the event structure from scratch, transforming it into a dynamic, business-oriented experience.
- **Event Backend Development**: Automated workflows for an 8-hour event managing 100+ Excel sheets, built using Python and AppScript.
- **Technical Leadership**: Led the technical execution of a large-scale event with **5,000+ applicants** and 200+ offline participants, automating workflows and real-time data processing for a **₹1L** prize pool.

CERTIFICATIONS

Machine Learning A-Z: AI, Python & R | Complete Python Developer Course | IBM DevOps & Agile Designs